
A/B Test Analysis of Subscription Conversion Optimization

Discount Messaging Impact on User Conversion

Problem Definition

Mobile app subscription paywall conversion rate is low

Current conversion rate: **6.10% (Group A baseline)**

Goal: Increase subscription purchases after onboarding

Hypothesis: Messaging strategy may impact user behavior

2000 users

34% Reach Paywall:
680 Users

6% Conversion: 611 Users



Hypothesis

→ **Null Hypothesis (H₀):**

There is no difference in conversion rate between A and B groups.

→ **Alternative Hypothesis (H₁):**

Discount messaging ("50% off") increases conversion rate.

→ **Expected effect:**

Increase from ~6% to higher conversion rate in Group B

Experiment Design

Control Group (A):

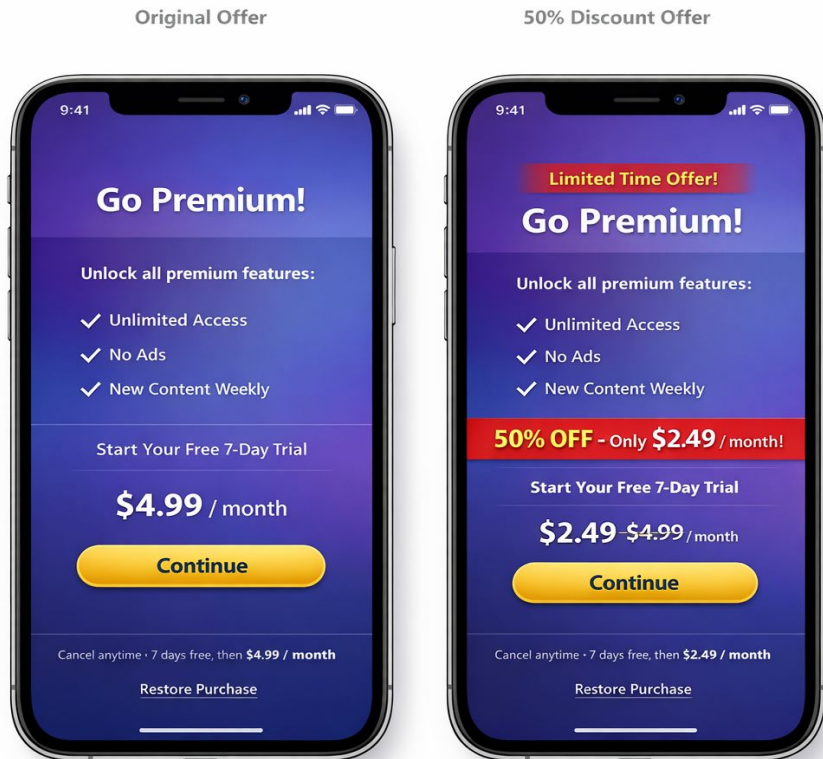
Standard \$4.99
subscription message

Test Group (B): "50%
discount" framing

Random assignment
(user-level)

Split: 50% / 50%

Unit: user_id



Metrics

Primary Metric:

Conversion Rate (Purchase
/ Paywall views)

Secondary Metrics:

Time to purchase

Payment success rate

Retention (optional)

Guardrails:

Crash rate

Sample ratio mismatch
(SRM)

Sample Size & Duration

- Daily users: 2000
- Paywall reach: 34% → **680 users/day**
- Per group: **340 users/day**

Required sample size: ~2626 users per group

Test duration: ~22 days

👉 Final duration used: **22 days**



2000 users/day

↓ **34%**

680 users

↓ ↓

340 A 340 B

↓

22 days

A/B Test Results

Group A:

Users: 10013

Conversions: 611

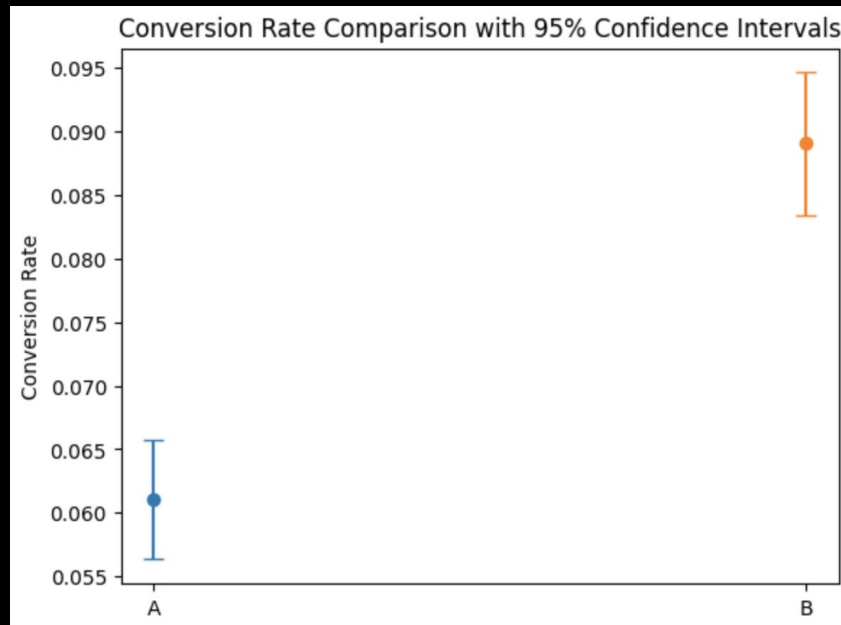
Conversion Rate: 6.10%

Group B:

Users: 9985

Conversions: 889

Conversion Rate: 8.90%



P-value: 0.0000

→ Statistically significant
difference ($p < 0.05$)



CONCLUSION

Conclusion & Decision

- Group B achieved a higher conversion rate than Group A
→ **8.90% vs 6.10%**
- Statistical test results:
→ **p-value ≈ 0.0000 ($p < 0.05$)**
→ Difference is statistically significant
- The “50% discount” message has a **positive impact** on user purchase behavior

Decision

👉 Roll out Group B to all users

Business Impact

- ~45% relative increase in conversion rate
- Higher subscription revenue potential
- Improved paywall performance